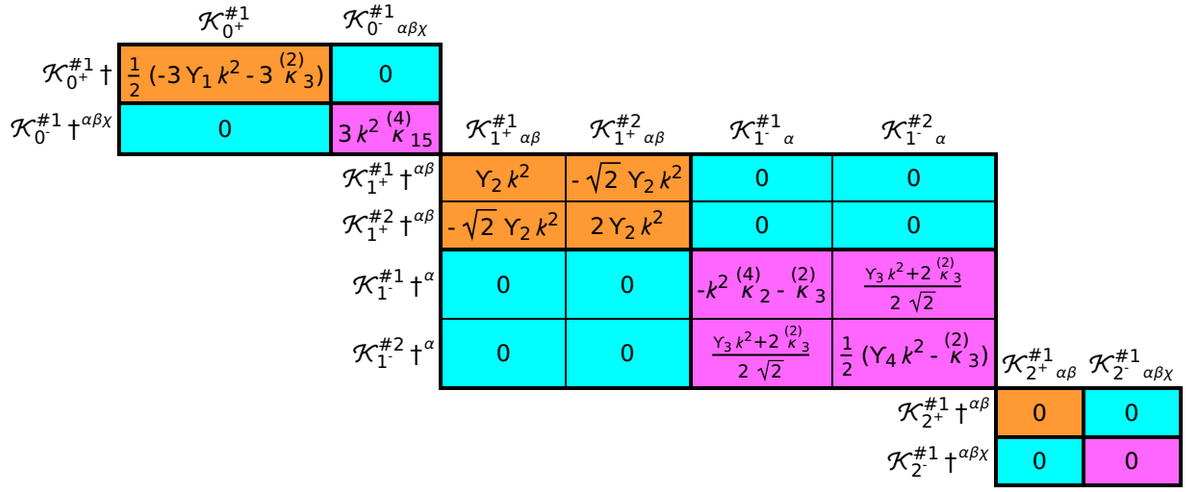
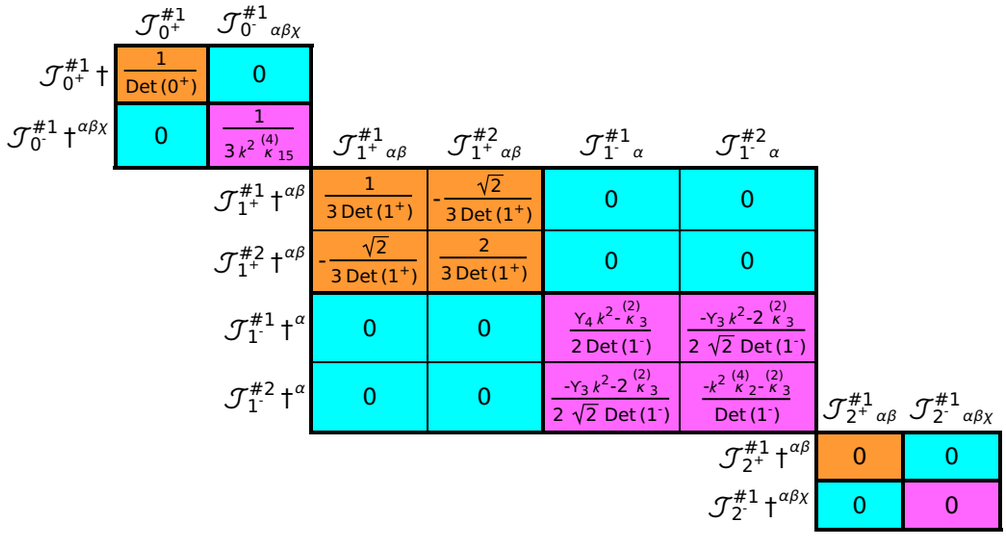




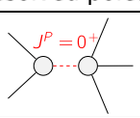
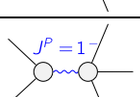
Abbreviations used in matrices

$$Y_1 == \overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2 \;\&\&\; Y_2 == 2\,\overset{(4)}{\kappa}_{15}-\overset{(4)}{\kappa}_3 \;\&\&\; Y_3 == 2\,\overset{(4)}{\kappa}_2+\overset{(4)}{\kappa}_7 \;\&\&$$
$$Y_4 == 18\,\overset{(4)}{\kappa}_{15}-\overset{(4)}{\kappa}_2-9\,\overset{(4)}{\kappa}_3-\overset{(4)}{\kappa}_7 \;\&\&\; \text{Det}(0^+) == -\frac{3}{2}(k^2(\overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2)+\overset{(2)}{\kappa}_3) \;\&\&\; \text{Det}(1^+) == 3\,k^2(2\,\overset{(4)}{\kappa}_{15}-\overset{(4)}{\kappa}_3)$$

Lagrangian

$$\overset{(2)}{\kappa}_3\,\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\mathcal{K}^{\chi}{}_{\beta\chi}+\overset{(4)}{\kappa}_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}+\overset{(4)}{\kappa}_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}+\overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta}-$$
$$12\,\overset{(4)}{\kappa}_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta}+4\,\overset{(4)}{\kappa}_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta}-\overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta}+\overset{(4)}{\kappa}_7\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta}+$$
$$3\,\overset{(4)}{\kappa}_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi}-2\,\overset{(4)}{\kappa}_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi}+\overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\alpha\beta\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}-2\,\overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\beta\alpha\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$2\,\mathcal{J}_{1^+}^{\#1\,\alpha\beta}+\mathcal{J}_{1^+}^{\#2\,\alpha\beta}==0$	3	$\partial_\chi\mathcal{J}^{\alpha\beta\chi}+\partial_\chi\mathcal{J}^{\chi\alpha\beta}==\partial_\chi\mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}_{2^-}^{\#1\,\alpha\beta\chi}==0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon}+3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta}+$ $4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi}+3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon}+3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon}+3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta}==$ $3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon}+3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta}+$ $2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}+3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon}+3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon}+3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$
$\mathcal{J}_{2^+}^{\#1\,\alpha\beta}==0$	5	$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta}+3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta}+2\,\eta^{\alpha\beta}\,\partial_\epsilon\partial^\epsilon\partial_\delta\mathcal{J}^{\chi}{}_{\chi}{}^{\delta}==2\,\partial_\delta\partial^\beta\partial^\alpha\mathcal{J}^{\chi}{}_{\chi}{}^{\delta}+3\,(\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}+\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi})$
Total # constraint(s):	13	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	$-\frac{\overset{(2)}{\kappa}_3}{\overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2}$	$-\frac{2}{3(\overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2)}$
	3	$\frac{36(-2\,\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_3)\,\overset{(2)}{\kappa}_3}{72\,\overset{(4)}{\kappa}_{15}\,\overset{(4)}{\kappa}_2-36\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3+\overset{(4)}{\kappa}_7{}^2}$	$\frac{432\,\overset{(4)}{\kappa}_{15}{}^2+108\,\overset{(4)}{\kappa}_3{}^2+12\,\overset{(4)}{\kappa}_3\,\overset{(4)}{\kappa}_7+\overset{(4)}{\kappa}_7{}^2-24\,\overset{(4)}{\kappa}_{15}(18\,\overset{(4)}{\kappa}_3+\overset{(4)}{\kappa}_7)}{3(2\,\overset{(4)}{\kappa}_{15}-\overset{(4)}{\kappa}_3)(72\,\overset{(4)}{\kappa}_{15}\,\overset{(4)}{\kappa}_2-36\,\overset{(4)}{\kappa}_2\,\overset{(4)}{\kappa}_3+\overset{(4)}{\kappa}_7{}^2)}$
Resolved unitarity condition(s):		(Demonstrably impossible)	